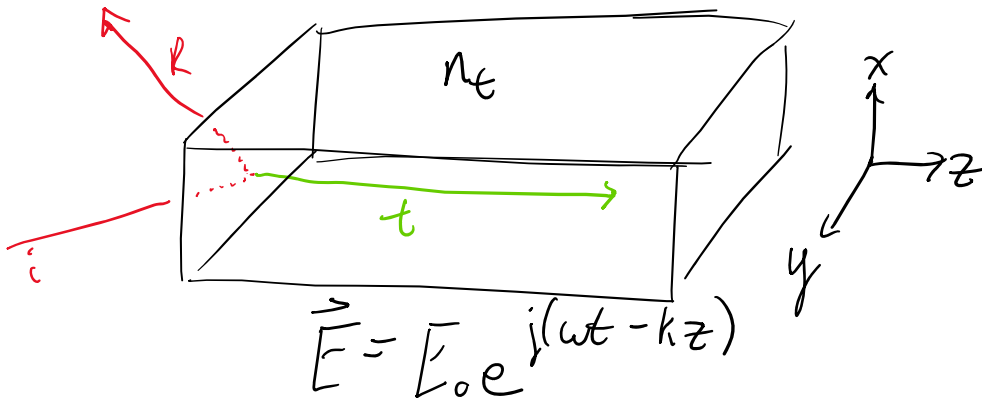


Losses in Dielectrics (Dielectric (ideally) has no free electrons)



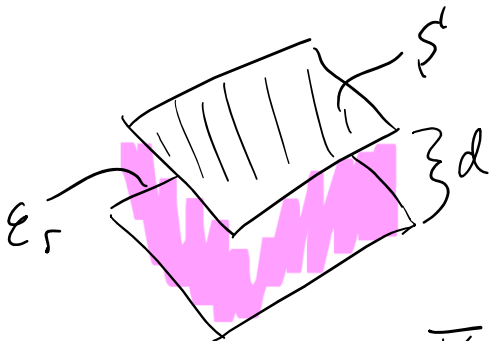
$$n_t = \sqrt{\epsilon_r}$$

Dielectric Loss

$$\epsilon_r = (\underbrace{\epsilon_r'}_{\text{storage term}} - j \underbrace{\epsilon_r''}_{\text{loss term}}) \rightarrow \text{complex relative permittivity}$$

the way to remember which term is loss is to think of a capacitor

$$C = \epsilon_0 \epsilon_r \frac{S}{d}$$



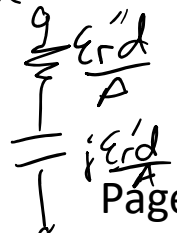
$$X_c = \frac{1}{j\omega C} = \frac{-j}{\omega \epsilon_0 \epsilon_r \left(\frac{S}{d}\right)} \rightarrow \frac{-jd}{\epsilon_0 \epsilon_r S \omega}$$

$$\epsilon_r = \epsilon_r' - j\epsilon_r''$$

$$X_c = \frac{-jd}{\omega \epsilon_0 S (\epsilon_r' - j\epsilon_r'')} \cdot \frac{(\epsilon_r' + j\epsilon_r'')}{(\epsilon_r' + j\epsilon_r'')}$$

$$X_c = \frac{-j\epsilon_r' d + \epsilon_r'' d}{\omega \epsilon_0 S ((\epsilon_r')^2 + (\epsilon_r'')^2)} = \underbrace{\left(\frac{\epsilon_r'' d}{A} - j \frac{\epsilon_r' d}{A} \right)}_{A}$$

ESR-equivalent series resistance



Dielectric Loss will affect the propagation constant (k)

$$k = \frac{2\pi}{\lambda_m} \quad \leftarrow \quad \lambda_m = \frac{c}{\sqrt{\epsilon_r} f}$$

wavelength
in the material

$$\lambda_0 = \frac{c}{f}$$

↑
free space
wavelength.

$$k = \frac{2\pi\sqrt{\epsilon_r}}{\lambda_0} \quad ; \quad \sqrt{\epsilon_r} \text{ is the refractive index}$$

Real life

$$k = \frac{2\pi\sqrt{(\epsilon_r' - j\epsilon_r'')}}{\lambda_0} \rightarrow \underline{k = k' - jk''}$$

$$E = E_0 e^{j\omega t} e^{-jkz}$$

$$E = E_0 e^{j\omega t} e^{-j(k' - jk'')z}$$

$$E = E_0 e^{-(k''z)} e^{j(\omega t - k'z)}$$



dielectric material that is really
long

- * water → Depth of the ocean. if black, no light
- * fiber optics.

$|E|^2$ proportional to Intensity (Π)

Π proportional to $e^{-(k''z)}$



$$\frac{d\Pi}{dz} = -2k''(\Pi)e^{-k''z}$$

$$2k'' = \alpha \text{ (attenuation)}$$

define a complex index of refraction (N)

$$N = (n - jK) = \left(\frac{k' - jk''}{k_0} \right) k_0 \text{ freespace prop. vector}$$

$$n = \frac{k'}{k_0} ; K = \frac{k''}{k_0}$$

extinction coefficient

refractive index

$$n - jK = \sqrt{\epsilon_r} = \sqrt{\epsilon_r' - j\epsilon_r''}$$

$$(n - jK)^2 = \epsilon_r' - j\epsilon_r''$$

$$(n - jK)(n - jK) = \epsilon_r' - j\epsilon_r''$$

$$n^2 - 2jKn - K^2 = \epsilon_r' - j\epsilon_r''$$

$$(n^2 - K^2) - j2Kn = \epsilon_r' - j\epsilon_r''$$

$$\epsilon_r' = n^2 - K^2$$

$$\epsilon_r'' = 2nK$$

Fresnel Reflection Coefficients are now

$$r = \frac{1 - n + jK}{1 + n - jK}$$

$$\tilde{R}(\text{Reflectance}) = \frac{|n - jK - 1|^2}{|n - jK + 1|^2} = \frac{(n-1)^2 + K^2}{(n+1)^2 + K^2}$$